React's component lifecycle is a crucial topic in interviews, as it plays a vital role in managing the behavior and performance of React applications. The lifecycle methods enable developers to perform operations like data fetching, resource cleanup, and DOM manipulation. By mastering React's component lifecycle, developers can build robust, efficient, and scalable applications. Interviewers often assess a candidate's understanding of these concepts to evaluate their ability to write effective, maintainable, and high-performance React code.

A.5.1 React class component lifecycle

React's component lifecycle is structured into three primary phases: Mounting, Updating, and Unmounting. While these phases apply to both class components and functional components, the implementation differs. Figure A.3 illustrates the lifecycle diagram for class components.

Figure A.3 Flowchart of React class component lifecycle phases, from mounting to unmounting, with associated methods